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Partial Differential Equations, Transforms & Optimization
Techniques

GROUP ASSIGNMENT & MICRO-PROJECT-1

PDE Applications in Engineering

Quantum Confinement (The Schrödinger Wave Equation)

*A Mathematical Derivation Using Separation of Variables and Fourier
Series*

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Contents

1	Objective	1
2	Introduction	1
3	Physical Model	2
4	Governing Equation	2
5	Separation of Variables	3
6	Elimination of Variables	3
7	Solution of the Temporal Equation	4
8	Solution of the Spatial Equation	4
9	Application of Boundary Conditions	5
9.1	Boundary Condition at $x = 0$	5
9.2	Boundary Condition at $x = L$	6
10	Quantized Energy States	7
11	Normalization of Eigenfunctions	7
12	Complete Stationary-State Solution	8
13	General Fourier Series Solution	9
14	Final Fourier Series Solution	10
15	Summary of Mathematical Results	10
16	Conclusion	11
17	References	12

1. Objective

The primary objective of this mathematical analysis is to derive the analytical solution of the one-dimensional time-dependent Schrödinger equation for an electron confined within an infinite potential well.

The specific objectives are:

1. To formulate the one-dimensional time-dependent Schrödinger equation for a particle inside an infinite potential well.
2. To apply the method of separation of variables and reduce the partial differential equation into independent ordinary differential equations.
3. To eliminate the variables and introduce the appropriate separation constant.
4. To solve the resulting spatial and temporal differential equations.
5. To apply the boundary conditions $\Psi(0, t) = \Psi(L, t) = 0$.
6. To determine the allowed wave numbers and demonstrate the quantization of energy.
7. To normalize the spatial eigenfunctions.
8. To construct the general solution using a Fourier series.
9. To obtain the Fourier coefficients from the initial wave function.

2. Introduction

The behavior of electrons and other quantum particles at nanoscale dimensions is governed by the principles of quantum mechanics. One of the fundamental equations describing such systems is the time-dependent Schrödinger equation.

In nanoscale electronic systems, particles may be confined to dimensions comparable to their quantum-mechanical wavelength. Under such conditions, the allowed energies are no longer continuous but occur as discrete energy levels.

A fundamental model for understanding this behavior is the one-dimensional infinite potential well. Although idealized, this model provides an important mathematical basis for understanding quantum confinement and energy quantization in nanoscale structures.

In this work, the one-dimensional time-dependent Schrödinger equation is solved analytically using separation of variables. The boundary conditions are then applied to obtain the allowed energy states, and the complete solution is represented as a Fourier series.

3. Physical Model

Consider a particle of mass m confined inside a one-dimensional infinite potential well of length L .

The potential energy function is

$$V(x) = \begin{cases} 0, & 0 < x < L, \\ \infty, & x \leq 0 \text{ or } x \geq L. \end{cases} \quad (1)$$

Inside the well,

$$V(x) = 0, \quad 0 < x < L. \quad (2)$$

Since the potential becomes infinite at the boundaries, the probability of finding the particle outside the well is zero. Therefore, the wave function must satisfy

$$\boxed{\Psi(0, t) = 0, \quad \Psi(L, t) = 0.} \quad (3)$$

4. Governing Equation

The general one-dimensional time-dependent Schrödinger equation is

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x) \Psi(x, t). \quad (4)$$

Inside the well, $V(x) = 0$. Hence,

$$\boxed{i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2}} \quad (5)$$

for

$$0 < x < L.$$

5. Separation of Variables

Assume that the wave function can be expressed as a product of a spatial function and a temporal function:

$$\boxed{\Psi(x, t) = \psi(x)\phi(t)} \quad (6)$$

The time derivative is

$$\frac{\partial \Psi}{\partial t} = \psi(x) \frac{d\phi}{dt}. \quad (7)$$

The second spatial derivative is

$$\frac{\partial^2 \Psi}{\partial x^2} = \phi(t) \frac{d^2 \psi}{dx^2}. \quad (8)$$

Substituting these expressions into Equation (5),

$$i\hbar\psi(x)\frac{d\phi}{dt} = -\frac{\hbar^2}{2m}\phi(t)\frac{d^2\psi}{dx^2}. \quad (9)$$

6. Elimination of Variables

Dividing both sides by $\psi(x)\phi(t)$ gives

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2\psi(x)}{dx^2}. \quad (10)$$

The left-hand side depends only on t , whereas the right-hand side depends only on x .

Therefore,

$$\text{function of } t = \text{function of } x = \text{constant}. \quad (11)$$

Let this separation constant be the total energy E .

Thus,

$$\boxed{i\hbar \frac{1}{\phi} \frac{d\phi}{dt} = E} \quad (12)$$

and

$$\boxed{-\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} = E.} \quad (13)$$

Hence, the original partial differential equation has been reduced to two ordinary differential equations.

7. Solution of the Temporal Equation

From Equation (12),

$$i\hbar \frac{d\phi}{dt} = E\phi. \quad (14)$$

Rearranging,

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar} \phi. \quad (15)$$

Separating variables,

$$\frac{d\phi}{\phi} = -\frac{iE}{\hbar} dt. \quad (16)$$

Integrating,

$$\ln \phi = -\frac{iEt}{\hbar} + C. \quad (17)$$

Therefore,

$$\phi(t) = C_1 e^{-iEt/\hbar}. \quad (18)$$

The constant C_1 can be incorporated into the spatial part. Thus,

$$\boxed{\phi(t) = e^{-iEt/\hbar}.} \quad (19)$$

8. Solution of the Spatial Equation

From Equation (13),

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi. \quad (20)$$

Rearranging,

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0. \quad (21)$$

Define

$$\boxed{k^2 = \frac{2mE}{\hbar^2}.} \quad (22)$$

Therefore,

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0. \quad (23)$$

The characteristic equation is

$$r^2 + k^2 = 0, \quad (24)$$

which gives

$$r = \pm ik. \quad (25)$$

Hence, the general solution is

$$\boxed{\psi(x) = A \sin(kx) + B \cos(kx).} \quad (26)$$

9. Application of Boundary Conditions

For the infinite potential well,

$$\boxed{\psi(0) = 0, \quad \psi(L) = 0.} \quad (27)$$

9.1 Boundary Condition at $x = 0$

Using Equation (26),

$$\psi(0) = A \sin(0) + B \cos(0) \quad (28)$$

$$= B. \quad (29)$$

Since $\psi(0) = 0$,

$$\boxed{B = 0}. \quad (30)$$

Therefore,

$$\psi(x) = A \sin(kx). \quad (31)$$

9.2 Boundary Condition at $x = L$

Applying

$$\psi(L) = 0,$$

we obtain

$$A \sin(kL) = 0. \quad (32)$$

For a non-trivial solution,

$$A \neq 0.$$

Therefore,

$$\sin(kL) = 0. \quad (33)$$

Hence,

$$kL = n\pi, \quad n = 1, 2, 3, \dots \quad (34)$$

and therefore,

$$\boxed{k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots} \quad (35)$$

Thus, the boundary conditions result in discrete allowed values of the wave number.

10. Quantized Energy States

From

$$k^2 = \frac{2mE}{\hbar^2}, \quad (36)$$

we obtain

$$E = \frac{\hbar^2 k^2}{2m}. \quad (37)$$

Substituting

$$k_n = \frac{n\pi}{L},$$

gives

$$E_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L} \right)^2 \quad (38)$$

$$= \frac{n^2 \pi^2 \hbar^2}{2mL^2}. \quad (39)$$

Therefore,

$$\boxed{E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots} \quad (40)$$

The energy spectrum is therefore discrete rather than continuous.

11. Normalization of Eigenfunctions

The spatial solution corresponding to the n th state is

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{L}\right). \quad (41)$$

For normalization,

$$\int_0^L |\psi_n(x)|^2 dx = 1. \quad (42)$$

Thus,

$$A^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1. \quad (43)$$

Using

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2}, \quad (44)$$

we get

$$A^2 \frac{L}{2} = 1. \quad (45)$$

Therefore,

$$A = \sqrt{\frac{2}{L}}. \quad (46)$$

Hence,

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right).} \quad (47)$$

12. Complete Stationary-State Solution

The complete wave function is

$$\Psi_n(x, t) = \psi_n(x) \phi_n(t). \quad (48)$$

Using Equations (47) and (19),

$$\Psi_n(x, t) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) e^{-iE_n t/\hbar}. \quad (49)$$

Substituting the quantized energy,

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad (50)$$

gives

$$\Psi_n(x, t) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \exp\left[-\frac{in^2 \pi^2 \hbar t}{2mL^2}\right]. \quad (51)$$

13. General Fourier Series Solution

The normalized eigenfunctions form an orthonormal basis:

$$\int_0^L \psi_n(x) \psi_m(x) dx = \delta_{nm}. \quad (52)$$

Therefore, an arbitrary initial wave function can be represented as

$$\Psi(x, 0) = \sum_{n=1}^{\infty} c_n \psi_n(x). \quad (53)$$

Substituting the normalized eigenfunctions,

$$\Psi(x, 0) = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right). \quad (54)$$

The Fourier coefficients are obtained from the orthogonality condition:

$$c_n = \sqrt{\frac{2}{L}} \int_0^L \Psi(x, 0) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (55)$$

Each eigenstate evolves with its corresponding phase factor $e^{-iE_n t/\hbar}$. Therefore,

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}. \quad (56)$$

Substituting the eigenfunctions,

$$\Psi(x, t) = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) e^{-iE_n t/\hbar}. \quad (57)$$

14. Final Fourier Series Solution

Using

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad (58)$$

the final general solution becomes

$$\Psi(x, t) = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) \exp\left[-\frac{in^2 \pi^2 \hbar t}{2mL^2}\right]. \quad (59)$$

where

$$c_n = \sqrt{\frac{2}{L}} \int_0^L \Psi(x, 0) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (60)$$

Equation (59) is the required general Fourier series solution.

15. Summary of Mathematical Results

The principal mathematical results obtained are

$$k_n = \left[\frac{n\pi}{L} \right], \quad (61)$$

$$E_n = \left[\frac{n^2\pi^2\hbar^2}{2mL^2} \right], \quad (62)$$

$$\psi_n(x) = \left[\sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \right], \quad (63)$$

$$\Psi_n(x, t) = \left[\sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) e^{-iE_n t/\hbar} \right], \quad (64)$$

$$c_n = \left[\sqrt{\frac{2}{L}} \int_0^L \Psi(x, 0) \sin\left(\frac{n\pi x}{L}\right) dx \right]. \quad (65)$$

Therefore,

$$\Psi(x, t) = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) \exp\left[-\frac{in^2\pi^2\hbar t}{2mL^2}\right]. \quad (66)$$

16. Conclusion

The one-dimensional time-dependent Schrödinger equation for a particle confined in an infinite potential well has been solved analytically using the method of separation of variables.

The separation

$$\Psi(x, t) = \psi(x)\phi(t)$$

reduces the original partial differential equation into independent spatial and temporal equations. Application of the boundary conditions

$$\psi(0) = \psi(L) = 0$$

restricts the allowed wave numbers to

$$k_n = \frac{n\pi}{L}.$$

Consequently, the energy of the particle becomes quantized:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}.$$

The corresponding normalized eigenfunctions are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right).$$

Finally, the general wave function is expressed as a Fourier superposition of these stationary states:

$$\Psi(x, t) = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) e^{-iE_n t/\hbar}.$$

Thus, the mathematical derivation demonstrates how the boundary conditions of an infinite potential well lead naturally to discrete energy states and how Fourier series provide the general solution for an arbitrary initial quantum state.

17. References

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